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The development of children's early memory skills

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ABSTRACT

A multitask battery tapping nonverbal memory and language skills was used to assess 60 children at 18, 24, and 30 months of age. Analyses focused on the degree to which language, working memory, and deliberate memory skills were linked concurrently to children's Elicited Imitation task performance and whether the patterns of association varied across the different ages. Language ability emerged as a predictor of *immediate* Elicited Imitation performance by 24 months of age and predicted *delayed* performance at each age. In addition to the contributions of language, children's abilities to search for and retrieve toys in the deliberate memory task were associated with their immediate Elicited Imitation performance at each age. In addition to language, working memory was positively associated with aspects of both immediate and delayed performance at all ages. The extent to which it was possible to replicate and extend previous cross-sectional work in this longitudinal study is discussed.

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Introduction

Young children's behavior reveals their abilities to remember long before they can use language to describe their experiences. Evidence from tasks that tap different aspects of nonverbal memory, including conditioning (e.g., Rovee-Collier & Hayne, 2000) and elicited and deferred imitation (e.g., Bauer, Wenner, Dropik, & Wewerka, 2000; Meltzoff, 1995), clearly documents the mnemonic competence of infants as well as substantial improvements with age in various aspects of remembering.

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Moreover, the literature on early memory that has emerged during the past 20 years has (a) contributed to a reappraisal of the cognitive competencies of infants, (b) suggested continuities across development in the operation of basic memory systems, (c) fostered explorations of critical constructs such as reinstatement, and (d) provided a foundation for addressing long-standing controversies such as infantile amnesia (see [Bauer, 2006](#), for a review).

Given these important contributions, as well as the fact that children's nonverbal memory skills may provide a platform for their later use of language to report the past (see [Bauer, 2007](#); [Ornstein, Haden, & San Souci, 2008](#)), it is surprising that our understanding of the development of nonverbal memory is limited by two features of the literature. First, because the literature on early memory is based largely on cross-sectional studies, it is difficult to move from statements about age-related changes in performance to descriptions of the developmental course of remembering within individual children. In this regard, longitudinal investigations in which the same children are traced over time are essential for describing how early skills are related to later abilities ([Ornstein & Haden, 2001, 2008](#)). Second, when children's memory—verbal or nonverbal—is assessed, the emphasis typically is on describing remembering in the context of one or another paradigm, with within-participant multitask assessment strategies being more the exception than the rule (but see [DeMarie & Ferron, 2003](#); [Lange & Carroll, 2003](#)). Because different assessments can result in quite different impressions of young children's memory skills ([Ornstein, Baker-Ward, & Naus, 1988](#); [Ornstein & Myers, 1996](#)), this is indeed unfortunate. Accurate characterization of children's changing skills requires that assessments include several tasks that vary in their information processing and effort requirements.

The current study is unique in coupling a multitask assessment strategy with a longitudinal research design in which children were tracked over the first 6 years of life. The focus in this report is on the memory and linguistic skills of 60 children (representing one of two cohorts that have been followed) who were assessed at three age points: 18, 24, and 30 months. The particular age points were selected because they span a period marked by the development and consolidation of children's memory as it has been assessed nonverbally along with the emergence of verbal recall. At each of the three age points, memory performance was examined in the context of three different nonverbal tasks that tap young children's changing event memory (Elicited Imitation [[Bauer et al., 2000](#)]), working memory (Working Memory for Locations [[Pelphrey & Reznick, 2003](#)]), and deliberate memory (Hide-and-Seek [[DeLoache, Cassidy, & Brown, 1985](#)]). Given the importance of examining developmental change in these skills, the primary aims of this investigation were to determine the extent to which children's abilities to reproduce modeled event sequences were associated *at each age* with their skills in language, deliberate memory, and working memory and to examine potential developmental changes in the linkages among these abilities.

The Elicited Imitation task was used to generate criterion measures because there is considerable agreement that toddlers' imitation of event sequences is based on memory processes that are comparable to the explicit memory skills revealed in older children's abilities to talk about past experiences and to prepare for deliberate assessments of remembering ([Bauer, 2006](#); [Ornstein et al., 2008](#)). In addition, there is a rich literature on age-related differences in children's memory for action sequences that have been modeled for them ([Bauer, 2007](#)), and considerably more is known about young children's performance on the Elicited Imitation task than on the Working Memory for Locations and Hide-and-Seek procedures. In the Elicited Imitation paradigm, remembering is demonstrated when a child uses props to reproduce a sequence of actions that had previously been modeled by an examiner. Consider, for example, the three actions involved in constructing a gong: putting a crossbar atop two posts, placing a metal plate on the crossbar, and then hitting the plate with a plastic mallet. After a baseline period, an experimenter demonstrates the actions required to construct the gong, typically with a verbal description of the behaviors and goal. Immediate and/or delayed memory for the modeled action sequences can be observed in children's responses to prompts such as "You show me how to make a gong." Using this method, [Bauer \(2007\)](#); see also [Bauer et al., 2000](#)) reported a number of age-related changes in early memory, including substantial increases in how long memories seem to last, and in children's abilities to evidence ordered recall after various delay intervals. In addition, with increases in age, children are better able to remember longer sequences of actions for greater periods of time. Perhaps most striking, the length of time across which ordered recall can be observed increases in a dramatic fashion during this time period; indeed, more than half of 20-month-olds are

able to reenact, in an ordered fashion, portions of to-be-remembered sequences after delays as long as 1 year (Bauer et al., 2000).

With the rich literature on age-related differences in children's memory for action sequences (Bauer, 2007) as a backdrop, the two primary questions addressed here were as follows. First, what skills *concurrently* predict Elicited Imitation task performance? Second, do these linkages vary with age? Few answers to these questions are available in the literature, although intriguing relations have been found between ratings of 20-month-olds' temperament, such as their abilities to sustain interest and focus attention, and corresponding individual differences in Elicited Imitation performance (Bauer, Burch, & Kleinknecht, 2002). Thinking of other skills that might contribute to variation in Elicited Imitation performance, it seems quite probable that children's basic memory capacity and deliberate memory skills—indexed by their performance on the Working Memory for Locations and Hide-and-Seek tasks, respectively—would be linked to their ability to remember previously modeled action sequences. Indeed, there is a substantial body of work suggesting that limitations in working memory can constrain subsequent cognitive operations and that age-related changes in the capacity of working memory have implications for understanding children's developing abilities in a range of cognitive tasks (Reznick, 2007). Moreover, recent analyses of deliberate and event memory tasks in terms of the same underlying factors that affect the encoding, storage, and retrieval of information (Ornstein, Haden, & Elischberger, 2006) would suggest similar linkages between early strategies and elicited imitation.

Therefore, it seems likely that performance on the Working Memory for Locations task should predict how well children can do on the Elicited Imitation task because working memory capacity may set fundamental limits on their abilities to represent and reproduce modeled action sequences. It also is possible that performance on a task in which strategies are activated—even the simple naming, pointing, and “peeking” behaviors observed by DeLoache and colleagues (1985) as objects are hidden—may also be predictive of how well children might do on the Elicited Imitation task because similar processes may underlie performance on each task. Thus, even though the assessment battery was designed to tap aspects of remembering that have been explored in quite different literatures (see Schneider & Bjorklund, 1998), there is good reason to expect across-task linkages.

An additional feature of this study is a consideration of children's language as a factor in their abilities to remember modeled action sequences. Language is thought to play a central role in the emergence and development of event memory as it is assessed verbally (see, e.g., Fivush, Haden, & Reese, 2006), but research involving imitation paradigms suggests that the effects of language might not be limited to verbal memory *per se* (e.g., Bauer & Wewerka, 1995; Hayne & Herbert, 2004; Herbert & Hayne, 2000). Although remembering in the Elicited Imitation task is reflected in the behavioral reproduction of specified actions, the modeling situation typically incorporates what can be viewed as a very helpful verbal narration of the episodes by the examiner. Thus, children have available to them at encoding not only the modeled actions but also a story that goes along with these actions, and children with advanced language skills may be better able to take advantage of the narrative than those with lower levels of language competency. Indeed, these children may be better able to attend to salient aspects of a modeled sequence, to make sense of the event as it unfolds, and to incorporate the examiner's narrative into their memory representations.

If language were to affect encoding of the action sequences in this manner, particularly when the children are 24 and 30 months of age, the outcome would be consistent with findings from a variety of studies suggesting that the specific language skills possessed by children at the time an event is experienced can influence the underlying representation and, thus, contribute to what can be later reported (e.g., Bauer, Kroupina, Schwade, Dropik, & Wewerka, 1998; Peterson & Rideout, 1998; Simcock & Hayne, 2002). Research conducted with imitation paradigms in particular suggests that verbal cues provided during the demonstration of an event and at the time of test can improve the nonverbal memory performance of children as young as 18 months (e.g., Hayne & Herbert, 2004; Herbert & Hayne, 2000). Admittedly, these linkages between children's language and nonverbal memory skills are not always obtained (see, e.g., Bauer et al., 2002, and Cheatham & Bauer, 2005, for contrasting results), but it nonetheless is important to examine these potential linkages with the same sample of children at multiple ages. Thus, in this study, children's language abilities were assessed with an

examiner-administered standardized instrument at the three target ages (Preschool Language Scale-3 [Zimmerman, Steiner, & Pond, 1992]).

In summary, the current study draws on longitudinal data obtained when children were 18, 24, and 30 months of age and makes use of an assessment battery that included three measures of their non-verbal memory and one measure of their receptive and productive language skills. It addresses questions concerning linkages at each age across nonverbal memory tasks and possible changes in the patterns of association as children's memory skills develop.

Method

Participants

Across two project sites (Chapel Hill/Durham, NC, and Chicago/Evanston, IL), 60 children (30 in North Carolina and 30 in Illinois, 33 girls and 27 boys) were initially enrolled in the study. The final sample was composed of 57 children, reflecting the loss of 3 participants (2 after the 18-month age point because their families moved and 1 after the 24-month age point because the family could not be contacted). On average, the children were 18 months 10 days of age (range = 17 months 21 days to 19 months 23 days) at the initial visit at the first assessment, 24 months 18 days of age at the second age point (range = 23 months 12 days to 25 months 25 days), and 30 months 21 days of age at the third age point (range = 29 months 14 days to 32 months 7 days). All children were from middle- to upper middle-class families. The sample was composed of 50 European Americans, 3 Asians, 2 African Americans, and 2 American Indians. At the 18-month age point, 12% of the children's mothers reported having had some college education, 50% reported having received a college degree, and 38% reported having earned an advanced degree.

Measures and procedures

At each age point, children were assessed three times in their homes. At 18 months, the first two visits were spaced approximately 2 weeks apart (mean delay between Visits 1 and 2 = 14.03 days), whereas at 24 and 30 months, the delay between the first and second visits was approximately 3 weeks (mean delay between Visits 1 and 2 = 21.12 days and 22.33 days at 24 and 30 months, respectively). These delay intervals were selected on the basis of a review of the Elicited Imitation task literature (e.g., Bauer et al., 2000) so as to achieve delays over which children at the specified ages would evidence recall along with some forgetting. The final visit at each age occurred approximately 1 day after Visit 2 (mean delays between Visits 2 and 3 = 1.1 days at 18 months, 1.2 days at 24 months, and 2.3 days at 30 months). As illustrated in Table 1, children participated in a range of tasks at each age point, and only those tasks in the battery that are pertinent to this article are described. One of four female graduate students (two at each research site) and two male graduate students (one at each research site) administered all of the tasks at a particular age, and often the same individual tested the children at repeated ages.

Elicited Imitation

Task description and procedure. The Elicited Imitation procedure was adapted from the work of Bauer and colleagues (2000) and involved using a set of props to model specific action sequences to each child individually. The number of steps in the sequence varied as a function of the child's age: three steps at 18 months, five steps at 24 months, and seven steps at 30 months. A complete list of the test sequences used is provided in Table 2. In addition to their length, the sequences also varied by type, such that half of the sequences were arbitrarily ordered (no inherent constraints on the temporal position of actions) and half involved only enabling (each action needed to be performed in a temporally invariant order to reach the end state) or a mix of enabling and arbitrary relations across the actions. The order of the presentation of the events was counterbalanced across children at each age point, and the same order was used at the immediate and delayed memory assessments.

Table 1

Timing of tasks administered to children at 18, 24, and 30 months.

Visit	18 months	24 months	30 months
1	Elicited Imitation: Modeling and immediate performance ^a	Elicited Imitation: Performance of 18-month events after long-term delay ^a Elicited Imitation: Modeling and immediate performance (new events) ^a	Elicited Imitation: Performance of 24-month events after long-term delay ^a Elicited Imitation: Modeling and immediate performance (new events) ^a
	Hide-and-Seek ^a Mother–child reminiscing Working Memory for Locations ^a	Hide-and-Seek ^a Mother–child reminiscing Working Memory for Locations ^a	Hide-and-Seek ^a Mother–child reminiscing
2	Elicited Imitation: Performance after delay ^a Preschool Language Scale-3: Receptive ^a	Elicited Imitation: Performance after delay ^a Preschool Language Scale-3: Receptive ^a	Elicited Imitation: Performance after delay ^a Preschool Language Scale-3: Receptive ^a
3	Story construction Working Memory Hide-and-Seek Preschool Language Scale-3: Expressive ^a	Story construction Working Memory Hide-and-Seek Preschool Language Scale-3: Expressive ^a	Story construction Working Memory Hide-and-Seek Preschool Language Scale-3: Expressive ^a
	Mother–child free play	Mother–child free play	Working Memory for Locations ^a Mother–child free play

Note. Tasks were administered within a session in the order they are listed in the table.

^a Task included in this report.

Table 2

Elicited imitation event sequences used.

Event and type	Description of action sequence
<i>Three-step action sequences performed at 18 months</i>	
Rattle (enabling)	Researcher modeled putting the block in one half of a barrel, putting the two halves of the barrel together, and shaking the barrel to make a noise
Gong (enabling)	Researcher modeled putting a crossbar atop the two posts, hanging a metal plate on the crossbar, and hitting the plate with a plastic mallet
Bunny (arbitrary)	Researcher modeled putting on the bunny's ears, putting on the bunny's eyes, and "feeding" the bunny a carrot (holding it to the bunny's mouth and saying "num, num, num")
Party hat (arbitrary)	Researcher modeled putting a cloth band around the bottom of the cone hat, putting a pom-pom on the top of the cone, and putting a sticker on the front of the cone
<i>Five-step action sequences performed at 24 months</i>	
Picnic (mixed)	Researcher modeled spreading out a small fabric square blanket, putting a bear on the blanket, putting a hat on the bear, cutting a birthday cake, and putting a piece of cake in a bowl
Train (mixed)	Researcher modeled putting together a train track, putting the train on the track, hanging a bell on the train, moving a roadblock, and driving the train under a bridge
Tugboat (arbitrary)	Researcher modeled putting the top on a tugboat, putting a life ring on the side of the tugboat, hooking up a raft to the tugboat, pulling in the anchor, and putting a light on top of the tugboat
Garden (arbitrary)	Researcher modeled raking a garden, watering the garden, putting a flower in a hole, attaching a garden gate, and hanging a sun on a post
<i>Seven-step action sequences performed at 30 months</i>	
Trucks (mixed)	Researcher modeled putting a seat in the truck, putting the driver in the seat, washing the tire, putting the screwdriver together, fixing the truck, loading the truck, and dumping it
Snow (mixed)	Researcher modeled pouring snowballs, making it snow, building a snowman, making a mountain, taking the bear skiing, pouring cocoa in the cup, and stirring the hot chocolate
Birthday art (arbitrary)	Researcher modeled decorating a piece of paper, rolling a paintbrush over the paper, putting on the decorative clip, putting on the frame, putting a bow on the frame, stamping the paper, and putting on a cookie sticker
School (arbitrary)	Researcher modeled driving a bus to school, hanging up the flag, sitting a child at the desk, putting up the chalkboard, playing on the see-saw, giving the teacher an apple, and erasing the chalkboard

Note. Types of events are in parentheses.

As suggested above, at each age point, the Elicited Imitation procedure involved an initial exposure and test period that was conducted at Visit 1 and a delayed recall assessment that took place at Visit 2. Prior to introduction of the new events at the 24- and 30-month age points, children's long-term memory for the events experienced at the previous age point (at 24 months, memory for the 18-month events; at 30 months, memory for the 24-month events; at 36 months, memory for the 30-month events) was also assessed. The Elicited Imitation task began with a familiarization period involving nontarget event props and actions that served to establish/reestablish researcher–child rapport and prop exchange routines (see Bauer et al., 2000, for details). At Visit 1 at each age point, familiarization was followed by the presentation of the four target event sequences. For each sequence in turn, the event-related props were first offered to the children to manipulate as they wished (e.g., “Look at this stuff,” “What can you do with this stuff?”) for a period of time that was dictated by each child's interest (i.e., it ended when the child pushed the props away or demonstrated some other sort of “off-task” behavior). Each child's spontaneous premodeling production of target actions with the props in the target order (e.g., for rattle, putting the block in the cup) formed the basis for determining his or her *baseline* performance.

Next, the specified action sequence was modeled two times in succession. During this modeling, the researcher labeled the sequence (e.g., “This is how I make a gong”) and then narrated the actions taken to produce the event. The period of exposure to the events was followed by two consecutive presentations of the props to the children when the researcher encouraged them to imitate that which had been modeled (e.g., “Can you make a gong just like I did?”). Children's postmodeling imitation of the events as modeled served as the basis for evaluating their *immediate* performance. After a delay of 2 weeks (at the 18-month assessment) or 3 weeks (at the 24- and 30-month assessments), Visit 2 began with a repetition of the familiarization period from Visit 1, followed by a re-presentation of each set of sequence-related props with no instruction or modeling. Children's use of these props to produce the events they had previously seen modeled was considered for evidence of *delayed* performance.

Scoring and measures. All sessions of this task were videotaped for later analysis. From the videotapes, at baseline and each memory test (immediate and delayed), two dependent variables were scored: (a) the total number of individual target actions produced (maxima = 3, 5, and 7 for each sequence on the three-, five-, and seven-step events, respectively) and (b) total number of pairs of actions produced in the target order (maxima = 2, 4, and 6 for each sequence on the three-, five, and seven-step events, respectively). For the sequencing measure, only the first occurrence of each target action was considered. For example, if a child produced all three actions in the three-step event in the target order, credit would be awarded for three target actions and two correctly ordered pairs of actions (pairing of Actions 1 and 2; pairing of Actions 2 and 3). However, if the child produced the first and third target actions in that order (i.e., Pair 1–3), 2 points would be credited for the production of target actions and 1 point would be credited for production of target sequences. Moreover, if the child produced the actions in the order Action 3–1–2–3, credit was awarded for the production of three target actions, but only 1 point was given for sequencing (i.e., Pair 1–2). Because Action 3 was initially performed prior to Actions 1 and 2, credit was not assigned for the correct ordering of the action when it was produced a second time.

Given that the primary focus of this study was on the prediction of Elicited Imitation task performance at each age and not on documenting the ubiquitous age-related increase in skill, the raw scores for each measure were selected for use in the analyses even though with increases in age the children were presented with events composed of different numbers of steps. Analysis involving direct comparisons across the different ages might require analysis of proportions rather than raw scores. In this regard, however, Bauer (e.g., Bauer et al., 2000) argued that analyses of Elicited Imitation proportion scores are problematic for the reason that it is not clear, for example, that recalling four of five actions, for a resulting proportion score of .80, is “better” than recalling two of three actions, for a proportion score of .67, because in each case the child is forgetting only one action of the sequence. Indeed, the use of proportions could potentially mask developmental changes in memory performance.

There were two coders at the 18- and 24-month assessments and three coders at the 30-month assessment, and at least one of these individuals participated in the scoring across adjacent age points. Interrater reliability was established in such a way that each rater at each age point coded 25% of the

baseline, immediate, and delay test sessions. Agreement between pairs of coders ranged from 84 to 100%, averaging 95, 95, and 94% at the 18-, 24-, and 30-month age points, respectively.

At each age point, children's demonstrations of memory for the target actions and pairs of actions in the Elicited Imitation task were averaged across types (arbitrary and enabling/mixed) to increase the generalizability of the findings across a range of task difficulties. Performance was summarized according to six different indexes: baseline, immediate, and delayed production of component actions and pairs of actions.

Hide-and-Seek

Task description and procedure. Following procedures detailed by DeLoache and colleagues (1985), each child accompanied the researcher as he or she hid a number of similarly sized familiar objects (e.g., a Big Bird doll, a ball) in several locations around a room in the child's house (e.g., under a pillow, behind a curtain). Then, following a delay period, the child was encouraged to search for each object until all objects were found. The task began with an initial familiarization trial in which only one object was hidden and retrieved immediately by the child. On the test trials that followed, the experimenter introduced two objects at 18 months, three objects at 24 months, and four objects at 30 months and indicated that the child should remember where these objects were hidden so that they could be retrieved later. Each child watched as the researcher hid all of the objects, and then a timer was set for 2 min at the 18-month assessment and for 3 min at the 24- and 30-month assessments. During the delay interval, the child was engaged in a coloring and/or block stacking activity in the center of the room, and when the timer sounded, the child was urged to find all of the objects. At each age point, this procedure was repeated for two trials. Although the same toys were used across trials, they were hidden in different locations in Trial 1 and Trial 2.

Scoring and measures. The analyses of this task focused exclusively on each child's memory performance, expressed as the total number of toys correctly retrieved from their hiding locations across the two trials regardless of the number of erroneous searches. Therefore, the total maximum scores were 4, 6, and 8, at the 18-, 24-, and 30-month assessment points, respectively. Based on the videotaped records, two research assistants established reliability for 25% of the sample at each age, and there was 100% agreement between the coders for each observation at each age point.

Working Memory for Locations

Task description and procedure. The procedure for this task was modeled after that used extensively in studies of working memory with infants and young children (e.g., Pelphrey & Reznick, 2003; Reznick, Corley, & Robinson, 1997). On each trial of this task, the child watched the researcher hide a small object under one of several inverted cups, after which the child's view of the cups was obstructed for a delay of a few seconds. Following the delay, the child was encouraged to find the object, and this procedure was repeated over a series of trials of increasing difficulty. At each age point, there was first a familiarization period during which the child was repeatedly presented with two cups at a 0-s delay until he or she demonstrated comprehension of the task of finding the hidden object. The first test trial involved two cups and a 1-s occlusion. Increases in the level of difficulty of subsequent trials involved first increasing the time the child was asked to wait to uncover the hidden object: two cups at 5 s and then two cups at 10 s. Next, the number of cups was increased to four, first with a 1-s delay and then with delays of 5 and 10 s. In principle, this sequence of delay intervals could have been repeated with six cups and delays up to 15 s, but not all children reached this level of difficulty. When the child failed to find a hidden object on a given trial (e.g., two cups and a 10-s delay), that same trial was repeated once, but a second error at the same level of difficulty marked the end of the task.

Scoring and measures. To evaluate the child's working memory capacity, a score was calculated based on the number of correct responses (i.e., the number of times the hidden object was uncovered). For example, a score of 4 indicated that the child had correctly retrieved the toy through four trials before making two consecutive errors; in other words, it indicated that the child had found the object under one of two cups after delays of 1, 5, and 10 s and under one of four cups after a 1-s delay. At each age point, there were two coders, and at least one of these individuals participated in the scoring across

adjacent age points. For each trial, the videotape records were reviewed and the coders scored the child's reaching behaviors as correct (i.e., uncovered the hidden object), incorrect (i.e., did not uncover the location where the object was hidden), or inappropriate (e.g., uncovered two locations at the same time). Reliability was established by two research assistants who coded 25% of the sample at each age, and agreement in making judgments on this task ranged from 80 to 100%, averaging 98, 99, and 99% at the 18-, 24-, and 30-month age points, respectively.

Language Skills

Task description and procedure. The Preschool Language Scale-3 (PLS-3; Zimmerman et al., 1992) was used to assess children's language skills at each age point. In general, the PLS-3 involves asking the child to point to pictures of named objects, describe pictured actions, and offer definitions of words and actions. The test was administered in the standard format, although to alleviate fatigue the Auditory Comprehension subscale was administered at Visit 2 and the Expressive Communication subscale was administered at Visit 3.

Scoring and measures. Raw scores for each subscale were computed according to the guidelines developed by the test's authors. These were converted to standard scores, and the sum of these standard scores was converted to a total language standard score for each child.

Results

Preliminary analyses

Preliminary analyses revealed that the 18-month baseline performance on the rattle event in the Elicited Imitation task was significantly higher than that on the other three events that were administered at that age ($F_s \geq 32.96$, $ps < .01$), suggesting that the children could essentially perform the activity even before it was modeled by the researcher. This being the case, the rattle event was excluded and the reported results are based on averages across the three remaining events at the 18-month assessment point, whereas at the two older ages the averages were computed across four events. The same procedure used to assess delayed performance after 2 or 3 weeks was used 6 months after modeling, but consistent with prior work (Bauer et al., 2000), preliminary analyses confirmed that Elicited Imitation performance did not differ between the 2- or 3-week and 6-month delays, although there was more missing data at the 6-month delay. Given the stability of performance over the short and long delay intervals, to avoid losing data, 6-month delayed scores were substituted for corresponding 2- or 3-week delay scores for the 2, 1, and 3 children who had missing 2- or 3-week delayed Elicited Imitation performance data at the 18-, 24-, and 30-month age points, respectively.

Descriptive analysis

The means and standard deviations of the four measures of Elicited Imitation task performance are presented in Table 3. At each age point, to determine whether the children learned the event sequences, their immediate and delayed imitation performance was compared with their baseline levels of performance. Immediate performance was also compared with delayed performance to gauge retention of information over time. To avoid redundancy with the regression analyses reported below, differences across ages were not tested in these analyses. Therefore, separate one-way analyses of variance (ANOVAs) with Bonferroni adjustment for multiple comparisons were conducted at 18, 24, and 30 months to test for effects of assessment (baseline, immediate, and delayed) on actions and pairs performance. These analyses yielded significant effects of assessment at each age for both actions ($F_s \geq 186.19$, $ps < .001$, $\eta^2s \geq .68$) and pairs ($F_s \geq 77.53$, $ps < .001$, $\eta^2s \geq .47$). As is illustrated in Table 3, at every age, the comparison of immediate performance with baseline performance indicated that the children had learned both the individual target actions and pairs of actions. Delayed performance was also better than baseline, although fewer actions and pairs were produced following the delay of weeks than immediately after modeling (all $ps < .001$).

Table 3

Means and standard deviations for children's Elicited Imitation task performance over time.

Elicited Imitation task performance	Age point		
	18 months	24 months	30 months
<i>Baseline</i>			
Actions	0.42 (0.38)	0.70 (0.47)	1.37 (0.65)
Pairs	0.02 (0.09)	0.11 (0.21)	0.30 (0.27)
<i>Immediate</i>			
Actions	2.34 (0.58)	4.34 (0.63)	5.75 (1.30)
Pairs	0.87 (0.47)	2.10 (0.58)	3.18 (0.99)
<i>Delay</i>			
Actions	1.77 (0.66)	3.25 (0.81)	4.56 (1.19)
Pairs	0.51 (0.44)	1.31 (0.58)	2.25 (0.81)

Note. Standard deviations are in parentheses. Performance is based on three-, five- and seven-step action sequences performed at 18, 24, and 30 months, respectively.

To further characterize the children's performance, the descriptive statistics for language skill as assessed via the PLS-3, for deliberate memory as measured in the Hide-and-Seek task, and for working memory as indexed in the Working Memory for Locations task are displayed in Table 4. The mean standard scores on the language measure suggest that the children were of average to above-average linguistic ability relative to the PLS-3 norming sample. A one-way ANOVA testing differences in total PLS-3 standard scores by age was statistically significant, $F(2, 162) = 18.09$, $p < .001$, $\eta^2 = .18$, and follow-up tests indicated that the children's 24- and 30-month language scores were significantly different from those at 18 months (the 24- and 30-month scores were not different from each other). The Hide-and-Seek measures indicate that the children were able to retrieve 2.10 (of 4) hidden toys at the 18-month assessment and 5.18 toys (of 8) by the time they were 30 months of age. Moreover, on the Working Memory for Locations task, the children could continue with the task through a mean of 3.05 trials (corresponding to finding an object underneath one of two cups after delays of 1, 5, and 10 s) when they were 18 months of age and 6.02 trials (finding an object located under one of four cups at a 10-s delay) by the time they were 30 months of age. One-way ANOVAs performed on these memory measures indicated significant improvements in performance at each age ($F_s = 70.12$ and 20.79, $\eta^2_s = .45$ and .19, for performance on the Hide-and-Seek and Working Memory for Locations tasks, respectively, $p_s < .01$).

Correlational analyses

A series of correlational analyses was conducted to examine (a) the stability of the children's performance over time and (b) the concurrent relations among the language and memory measures at each age point. Turning first to the issue of stability of performance, correlations were computed for performance on each measure between 18 and 24 months, between 18 and 30 months, and between 24 and 30 months. As illustrated in Table 5, although children's early and later language skills were interrelated, in general, for the memory measures little stability in the children's performance

Table 4

Means and standard deviations for language, deliberate memory, and working memory measures.

Measure	Age point		
	18 Months	24 Months	30 Months
Language	100.45 (15.36)	112.88 (16.34)	118.17 (15.38)
Deliberate memory	2.10 (1.12)	3.64 (1.49)	5.18 (1.57)
Working memory	3.05 (1.83)	4.59 (2.70)	6.02 (2.85)

Note. Standard deviations are in parentheses.

Table 5

Across-time correlations for memory and language measures.

Measure	Age point		
	18–24 months	18–30 months	24–30 months
<i>Elicited imitation</i>			
Immediate actions	.35**	.29*	.23
Immediate pairs	.26*	.19	.26*
Delayed actions	.38**	.28*	.43***
Delayed pairs	.07	.13	.28*
Language	.54***	.46**	.69***
Deliberate memory	.27*	.08	.21
Working memory	.19	.25	.24

* $p < .05$.** $p < .01$.*** $p < .001$.

was found across the different ages. In particular, looking at the Elicited Imitation task measures, the children's performance of actions immediately after modeling at 18 months was modestly associated with immediate performance of actions at the later two time points. For the immediate pairs measure, fairly low associations were found in children's performance between 18 and 24 months and between 24 and 30 months. For the delay measures, stability was found in performance of actions between 18 and 24 months, between 18 and 30 months, and between 24 and 30 months, but only in performance of pairs of actions between 24 and 30 months. With regard to deliberate memory and working memory, the only across-time correlation found was a small positive association for children's performance on the deliberate memory task between the 18- and 24-month age points.

In the concurrent analyses of interrelations among the memory and language measures, deliberate memory was not significantly correlated with working memory ($r_s = .11$ – $.20$, $p_s > .13$) or with language skills ($r_s = .08$ – $.21$, $p_s > .13$) at any age. Working memory was linked to language skills at 24 months ($r = .29$, $p < .05$), but not at 18 or 30 months ($r_s = .06$ and $.11$, respectively, $p_s > .13$). The question of whether the children's Elicited Imitation task performance at each age was concurrently related to their language, deliberate memory, and working memory skills was addressed in the main analyses that are summarized below.

Concurrent prediction of Elicited Imitation task performance at each age

To examine the extent to which the language and memory measures were linked concurrently to the children's performance on the Elicited Imitation task, repeated measures analyses using a general linear mixed model approach were conducted on the 18-, 24-, and 30-month imitation indexes twice for each retention interval (i.e., immediate and delay). In each analysis, the repeated measures were accounted for through an autocorrelation structure that allowed assessments closer in time to be more highly correlated. The predictors included the repeated measures of language, working memory, and deliberate memory, also collected at 18, 24, and 30 months of age, as time-varying covariates.

Several models were fit to predict the data for each of four indexes of Elicited Imitation task performance: immediate and delayed production of component actions and pairs of actions. The models predicting delay performance were run twice: once without controlling for the children's immediate performance and a second time with immediate performance as a covariate. The first model for the prediction of each index included gender and age and tested the extent to which each index varied over time and whether these patterns were different for boys and girls; site was also included in the first model to examine the comparability of findings across our two test sites. The second model added language skills prior to the deliberate memory and working memory measures because, based on previous research, it could be expected that concurrent linguistic skills would be associated with the children's Elicited Imitation performance. The third step involved two separate models, 3A and 3B, for each index of Elicited Imitation performance. Model 3A built on the second model by adding concurrent deliberate

memory performance and its interaction with age to determine the extent to which deliberate memory predicted Elicited Imitation performance and whether the prediction varied as a function of age. In a parallel fashion, Model 3B extended the second model by adding the time-varying measures of working memory instead of those of deliberate memory performance. In the absence of a priori evidence concerning the relative importance of deliberate and working memory skills as predictors of Elicited Imitation performance, these two separate but parallel models were established to allow independent assessments of the contribution of each of these two predictors as opposed to adding one before the other or using a single model that included both measures simultaneously.

In the second and third steps (both Models 3A and 3B) of these analyses, the main effect of each predictor (i.e., language, deliberate memory, and working memory) describes an association with an index of Elicited Imitation task performance and, in the absence of an interaction with age, suggests that a similar concurrent association between the predictor and Elicited Imitation outcome exists at all ages. The Predictor $\times$ Age interactions describe associations with Elicited Imitation performance that change over time; it should be noted that at each step, nonsignificant interactions were dropped from the model. The results of these analyses are summarized in Table 6. In addition to the F values and betas displayed in this table, effect sizes (d values) were calculated to describe the magnitude of the associations among the Elicited Imitation and predictor variables. Each effect size was computed as the regression coefficient multiplied by the standard deviation of the predictor variable (i.e., language, working memory, and deliberate memory) divided by the standard deviation of the Elicited Imitation performance measure. For these effect sizes, d values below .20 are traditionally viewed as modest, d values between .20 and .60 are taken to be moderate, and d values above .60 are seen as large (Cohen, 1988).

The upper section of Table 6 displays the results for the initial step or base model that characterizes changes in Elicited Imitation task performance as a function of age or time of assessment and includes research site and gender as covariates along with the Gender $\times$ Age interaction that was computed to examine possible differences in patterns of change over time for boys and girls. The middle portion of the table presents the second step and illustrates the effects of adding language and the Age $\times$ Language interaction to the base model, and the bottom portion shows the two third-step models that evaluate the additional impact of the deliberate and working memory measures and their interactions with age as predictors of the children's production of modeled action sequences. The test statistics are listed for each main effect and for the interactions with age. In addition, the adjusted means and their standard errors are shown for the categorical predictor, age, and regression coefficients, with the standard errors being displayed for each continuous predictor. The results for each part of the analyses are presented below.

Base model

Inspection of the base model results presented in the top portion of Table 6 suggests that all four measures of the children's Elicited Imitation task performance were comparable at the two research sites. There were, however, main effects of gender and age. Specifically, the tests involving gender indicate that girls outperformed boys on all measures of Elicited Imitation performance. Comparison of the analyses of delayed performance without and with immediate performance as a covariate suggests that the effects of gender at all ages were due primarily to the higher production of actions and pairs at the immediate assessment by the girls, in contrast to the boys. The girls also produced more actions than the boys at the delayed assessment at the 30-month age point.

The main effects of age reflect differences in the number of actions and pairs of actions produced at each age point, with the adjusted mean scores in Table 6 illustrating higher levels of initial and delayed performance at the 24- and 30-month age points relative to that observed at the 18-month assessment. When considering the main effects of age for Elicited Imitation task performance, it is important to keep in mind that the children were performing events with different numbers of steps at each age.

Language, deliberate memory, and working memory

To address the question of the extent to which the children's Elicited Imitation task performance is associated with their language abilities, skills for deliberate remembering, and working memory

Table 6

Predicting Elicited Imitation task performance from language, deliberate memory, and working memory skill.

Predictor		Elicited imitation task performance											
		Immediate				Delay				Delay (covarying immediate performance)			
		Actions		Pairs		Actions		Pairs		Actions		Pairs	
<i>Step 1: Base model</i>													
Site	F	0.02		0.00		1.27		0.08		2.74		0.03	
Gender	F	4.86 [*]		4.77 [*]		5.00 [*]		4.92 [*]		2.41		1.67	
Age	F	226.21 ^{***}		158.08 ^{***}		163.76 ^{***}		116.78 ^{***}		0.42		1.83	
18 months	M ^a (SE)	2.31	(0.12)	0.86	(0.09)	1.75	(0.12)	0.50	(0.08)	2.84	(0.29)	0.92	(0.18)
24 months	M ^a (SE)	4.32	(0.12)	2.09	(0.09)	3.24	(0.12)	1.31	(0.08)	3.09	(0.09)	1.28	(0.07)
30 months	M ^a (SE)	5.74	(0.12)	3.16	(0.10)	4.52	(0.12)	2.22	(0.08)	3.42	(0.14)	1.67	(0.01)
Gender × Age	F	0.22		0.71		1.44		2.20		3.19 [*]		1.70	
Gender at 18 months	B (SE)									.015	(.182)		
Gender at 24 months	B (SE)									-.022	(.174)		
Gender at 30 months	B (SE)									.515 ⁺	(.172)		
Immediate (covariate)	F									83.81 ^{***}		52.37 ^{***}	
Immediate (covariate) × Age	F									0.32		0.81	
<i>Step 2: Language</i>													
Language	F	17.02 ^{**}		13.77 ^{***}		16.53 ^{***}		14.35 ^{***}		4.42 [*]		3.65 ⁺	
Language	B (SE)	.020 ^{***}	(.005)	.015 ^{***}	(.004)	.020 ^{***}	(.005)	.013 ^{***}	(.003)	.008 [*]	(.004)	.006 ⁺	(.003)
Language × Age	F	6.00 ^{**}		4.15 ⁺		1.18		0.45		0.92		0.31	
Language at 18 months	B (SE)	.006	(.007)	.003	(.006)								
Language at 24 months	B (SE)	.011	(.007)	.012 [*]	(.006)								
Language at 30 months	B (SE)	.042 ^{***}	(.008)	.029 ^{***}	(.070)								
<i>Step 3A: Deliberate memory</i>													
Deliberate memory	F	11.59 ^{**}		14.01 ^{***}		6.37 [*]		2.82 [*]		0.57		0.01	
Deliberate memory	B (SE)	.166 ^{***}	(.048)	.149 ^{***}	(.040)	.122 [*]	(.048)	.060 ⁺	(.035)	.030	(.040)	.003	(.033)
Deliberate Memory × Age	F	1.33		1.81		0.79		0.78		0.02		0.20	
<i>Step 3B: Working memory</i>													
Working memory	F	8.18 ^{**}		5.31 ⁺		15.35 ^{***}		10.62 ^{**}		11.42 ^{**}		6.23 [*]	
Working memory	B (SE)	.081 ^{**}	(.029)	.055 ⁺	(.024)	.108 ^{***}	(.027)	.065 ^{**}	(.020)	.078 ^{**}	(.023)	.045 ⁺	(.018)
Working Memory × Age	F	1.39		1.01		2.41 ⁺		1.93		2.82 [*]		3.76 [*]	
Working memory at 18 months	B (SE)					.194 ^{**}	(.058)			.156 ^{**}	(.052)	.047	(.040)
Working memory at 24 months	B (SE)					.051	(.038)			.019	(.031)	-.003	(.025)
Working memory at 30 months	B (SE)					.079 ⁺	(.037)			.060 ⁺	(.029)	.091 ^{***}	(.024)

^a Means are adjusted for the covariates: site and gender.* $p < .10$.* $p < .05$.** $p < .01$.*** $p < .001$.

capacity, as indicated above, one repeated measures analysis model in which language was added to the base model was carried out (see Step 2 in Table 6). Next, two additional parallel conditional models were tested, one adding deliberate memory and the other adding working memory, once the effects of language had been taken into account (see Steps 3A and 3B in Table 6). Given that deliberate memory and working memory skills, as assessed, were not interrelated, this approach was taken to test how each skill might contribute independently to children's abilities to reproduce modeled event sequences.

Language

As illustrated in the second section of Table 6, the children's total language standard score from the Preschool Language Scale-3 was significantly associated with their immediate production of component actions and pairs of actions in the Elicited Imitation task, and these associations, although modest, become stronger with increasing ages. Specifically, the Language $\times$ Age interactions with immediate performance indicate that language skills emerge as a predictor of the production of pairs of actions at 24 months and of individual components as well as pairs at 30 months. For example, at 30 months, the children increased their immediate production of actions by .042 for every 1-point increase in their Preschool Language Scale-3 total score. For the association between language and immediate performance of actions, the effect size ranged from $d = .16$ at 18 months, to $d = .34$ at 24 months, to $d = .50$ at 30 months, whereas for the relation between language and the immediate production of pairs of actions, it ranged from $d = .10$ at 18 months, to $d = .28$ at 24 months, to $d = .45$ at 30 months.

In contrast to immediate Elicited Imitation task performance, assessment of the children's performance following the delay (illustrated in the middle and right-hand sets of columns in Table 6) indicates main effects of language in the absence of Language $\times$ Age interactions. These outcomes suggest that the children's language skills were positively associated with delayed performance of both the actions and pairs of actions at each age. The effect sizes associated with these main effects were $d = .24$ for the components measure and $d = .24$ for the pairs measure without controlling for immediate performance. It is noteworthy that when the children's immediate performance was included as a covariate, the main effects of language on delayed performance were reduced in magnitude, still statistically significant for the production of actions and reflecting a statistical trend for the production of pairs. These results suggest that beyond both the children's immediate performance and the association between immediate performance and their language skills, language abilities contribute uniquely, albeit modestly, to delayed Elicited Imitation performance at all three ages.

Memory skills

As indicated in the third section of Table 6 (Step 3A), over and above the effects of language, there is evidence for the significant unique contributions of deliberate memory skill to the prediction of children's immediate Elicited Imitation task performance for both the components and pairs measures. Effect sizes of $ds = .19$ and $.24$ were obtained for the immediate actions and sequencing measures, respectively. Thus, children's abilities to search and retrieve toys in the Hide-and-Seek task were associated with their immediate postmodeling production of the actions and ordering of the actions in the Elicited Imitation task. Moreover, the failure to obtain Deliberate Memory $\times$ Age interactions suggests that these associations did not vary significantly with age. Comparison of the analysis involving delayed performance without and with controlling for immediate performance further suggests that deliberate memory skill was primarily impacting the children's immediate performance. More specifically, although without controlling for immediate performance, performance on the deliberate memory task was significantly associated with delayed production of components and tended to be associated with the delayed production of pairs, with effect sizes of $ds = .16$ and $.12$, respectively, these results were reduced to nonsignificance when immediate performance was included as a covariate.

In addition to the children's language skills, their working memory capacity as assessed on the Working Memory for Locations task was also found to be positively associated with immediate and delayed performance of both actions and correct ordering of pairs of actions on the Elicited Imitation task. The effect sizes associated with these linkages were $ds = .13$ and $.13$ for immediate performance and $ds = .20$ and $.19$ for delayed performance of actions and pairs, respectively. Furthermore, a signif-

icant Working Memory $\times$ Age interaction for the delayed production of pairs of actions was obtained once immediate performance was controlled. This interaction suggests that at 30 months, the higher the children's working memory skills the higher their performance at sequencing of target actions after a 3-week delay. For the association between working memory and the delayed production of pairs, the effect sizes ranged from $d = .21$ at 18 months, to $d = -.01$ at 24 months, to $d = .32$ at 30 months. More specifically, at 30 months, for every increase of 1 point in working memory capacity, there was a corresponding increase in performance of .091 action pairs.

Discussion

This investigation was motivated by two salient gaps in the literature concerning children's early memory skills. First, despite innovative research illustrating the impressive mnemonic competencies that begin during infancy and continue through toddlerhood, little is known about how different measures of young children's explicit memory may be interrelated, particularly those tapping nonverbal aspects of remembering. Second, because few studies have tracked the same children over time, there is a paucity of information concerning the changing patterns of association across tasks that may be observed with development. Capitalizing on the longitudinal design employed in the current study, it was possible to establish at each of three age points the extent to which working memory, deliberate memory, and language predict the children's Elicited Imitation task performance as well as whether the patterns of association are different across the different ages.

To a considerable extent, the average levels of performance that were found on each of the memory tasks replicate prior work in which one or another of these nonverbal memory tasks has been employed with cross-sectional samples. Indeed, with few exceptions, the mean levels of memory demonstrated on each task were very similar to values that have been reported in previous research for children of similar ages. Consider, for example, that in a study where children were exposed to three-step action sequences six times across three exposure sessions, [Bauer and colleagues \(2000\)](#) found that 16-month-olds' productions of actions and pairs of actions ranged from 2.59 to 2.94 and from 1.11 to 1.44, respectively. In comparison, even though the children in the current study were slightly older and received only two demonstrations of each three-step action sequence, their immediate performance was just slightly lower than that reported by Bauer and colleagues, with an average of 2.34 component actions and 0.87 action pairs produced at 18 months. It is also worth noting that the advantage for girls found in this study is consistent with the only other report in the literature to find effects of gender on this task with children 2 years of age or older ([Bauer, Hertsgaard, Dropik, & Daly, 1998](#)).

Likewise, whereas [DeLoache and colleagues \(1985, Experiment 4\)](#) demonstrated that 21- to 25-month-olds retrieved approximately 53% of the toys that were hidden in a laboratory setting, the average retrievals for the current sample at 24 months was a very comparable 60% or 3.64 toys retrieved from locations around the family's home. Although it is difficult to find working memory data in the literature that map neatly onto the task parameters employed here, the general sense is one of comparability. For example, the performance of the children in the current study is just shy of that reported by [Reznick and colleagues \(1997\)](#) in a longitudinal twin study. To illustrate, Reznick and colleagues observed that 20-month-olds could retrieve a toy that had been hidden beneath one of six cups after a delay of 1 s, whereas at 18 months the children in the current sample could retrieve a toy that was hidden under one of four cups after a 10-s delay.

Despite the focus on nonverbal assessment of young children's remembering in this investigation, the first step in the analyses was to consider how children's language proficiency at the time of the experience of the Elicited Imitation task events might be associated with their performance. Admittedly, as in studies of categorization and theory of mind that have focused on the relations between children's language and other areas of cognitive development (see, e.g., [Astington & Baird, 2005](#), on theory of mind; [Rakison & Oakes, 2003](#), on categorization), the extant literature on children's memory is mixed, with reports of both linkages between children's language skills at encoding and subsequent remembering and of failures to find these associations. Interestingly, these inconsistent findings can even be observed within the same laboratory with the same assessment tool, the MacArthur

Communicative Development Inventories (cf. Bauer et al., 1998, 2002; Cheatham & Bauer, 2005). In the current investigation, it was observed that children's language skills were associated with performance immediately after modeling for both the actions and pairs measures of memory in the Elicited Imitation task. Thus, to a considerable extent, these results are similar to and extend those obtained by Bauer and colleagues (1998) and Bauer and Wewerka (1995), in which the productive vocabulary of 16- and 20-month-olds was related, albeit modestly, to memories expressed nonverbally at an immediate assessment. Accordingly, the current findings suggest the tentative conclusion that at least by 2 years of age, the language available to children as an event unfolds is an important determinant of what gets into memory (see also Boland, Haden, & Ornstein, 2003).

Somewhat in contrast to Bauer and colleagues' (1998) and Bauer and Wewerka's (1995) studies, however, the language skills of the children in the current study were related to nonverbal measures of performance after a 2- or 3-week delay. Indeed, that this effect was maintained even after controlling for immediate Elicited Imitation task performance suggests that linguistic skills can exert unique effects on children's remembering (see also Hayne & Herbert, 2004). Previous studies document substantial linkages between children's linguistic abilities after an event has been experienced and subsequent verbal indications of long-term retention and forgetting (see Bauer, 2006; Fivush et al., 2006), and the present results for nonverbal production are in accord with these findings. However, it must be emphasized that the current sample had generally high standardized language scores, and future work is needed to address what linkages might be found with a more linguistically diverse group. What is more, to say that there is a relation between language and children's verbal or nonverbal event memory is to oversimplify, and the critical question for future research is *why* language matters for children's developing skills for remembering.

Over and above the associations with language, measures of the children's immediate production of modeled sequences were associated with their deliberate memory performance, and these linkages were observed at all three ages. Importantly, however, the observation that the associations between the Elicited Imitation task performance and deliberate memory skill as assessed on the Hide-and-Seek task were lost once immediate performance was controlled suggests that the linkage between Elicited Imitation performance and deliberate memory skill is principally at the level of learning the target actions and sequences of actions. This finding may reflect the operation of similar factors that impact the children's encoding and the reproduction of an adult's actions after a short period of time regardless of whether they are explicitly told to remember the researcher's activity of hiding objects, as in the Hide-and-Seek task, or they are engaged in a game where the rule is to "make an X" in just the way it was modeled by the researcher, as in the Elicited Imitation task. For example, successful performance on each task may involve the children's ability to focus attention on the researcher's actions at the time of initial exposure and also, perhaps, a sensitivity to the need to do something in an effort to remember.

These across-task relations on nonverbal event and deliberate memory tasks are particularly interesting when contrasted with the mixed findings regarding preschoolers' and older children's performance on verbally based tasks. For example, Haden, Ornstein, Eckerman, and Didow (2001) reported relatively strong associations between children's event recall at 30 months and their deliberate memory performance as much as a year later, whereas Lange and Carroll (2003) obtained negative across-task correlations. Although these differences could stem from the substantially different tasks and methodologies used to assess children's event and deliberate memory skills and the longitudinal versus cross-sectional analytic strategies employed in these two studies, they may also reflect the dramatic transitions in cognition that are observed during the preschool period. Consistent with this interpretation, studies involving school-aged children suggest few linkages across verbal memory tasks (Knopf, Körkel, Schneider, & Weinert, 1988; Weinert & Schneider, 1995) and even dissociations among different measures of children's deliberate memory skills. It seems clear that further research is needed to understand the extent to which children who have better memories for events also prove to be more skilled—concurrently as well as longitudinally—at both nonverbal and verbally based deliberate memory tasks.

In addition to the contributions of language skills to the concurrent Elicited Imitation task performance, the children's working memory capacity was also positively associated with immediate performance of modeled actions and with the delayed production of both actions and action pairs. Moreover, with immediate performance controlled, it was apparent that by 30 months, the better the children's

working memory skills, the better their delayed production of sequences of pairs of actions. These findings are consistent with the view (Ornstein et al., 2006; Reznick, 2007) that the capacity of working memory sets limits on the amount of information that can be encoded and subsequently retrieved. Indeed, the linkage between performance on the Working Memory for Locations task and delayed Elicited Imitation task performance may reflect the quality of the representation that was established initially. To be sure, within the context of the current longitudinal investigation, these data set the stage for further exploration of the ways in which basic memory capacity interacts with a range of child and contextual variables to influence the development of a broad set of skills for remembering.

The current study is unique in its demonstration of concurrent relations among measures of children's language, working memory, deliberate memory, and event memory at multiple age points as these skills emerge and become consolidated. Nevertheless, additional work is clearly necessary to replicate and extend these findings with longitudinal research that includes assessments of children's performance on multiple tasks. It is particularly important to specify key underlying mechanisms for across-task associations and to explore the extent to which, for example, there may be similar processes at play as children encode events that they are experiencing and "study" items that have been hidden (Ornstein et al., 2006). In addition, given the broad longitudinal investigation in which this study is embedded, the current data can offer a foundation for future research addressing the prediction of children's later skill in verbally based assessments of remembering, both those involving the incidental recall of personally experienced events and those calling for deliberate preparation for subsequent assessments of memory. Although it is difficult to talk about the prediction of skills that are undergoing rapid development, it seems likely that composite measures of children's skill in the 18- to 30-month period will be reliably linked to performance during the preschool and early school years.

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